

DSA 210 — Introduction to Data Science

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Personalized Well-being Prediction from Sleep & Lifestyle Data

Final Project Report

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Introduction

Sleep is one of the most significant determinants of cognitive performance, mood, and physical health. Despite this, its personal impact at the individual level is poorly understood: commercial sleep trackers provide raw sensor readings without personalised interpretation or actionable insight. This project addresses that gap by building a dual-target predictive model trained on 44 consecutive nights of self-collected personal data.

The two prediction targets are the **morning feel rating** — how rested I feel upon waking, on a 0–10 scale — and the **daily productivity rating** — how productive my following day turns out to be, also on a 0–10 scale, logged each evening. A secondary objective is to derive a personalised daily routine recommendation from the model’s feature importance outputs, identifying the conditions under which both outcomes are jointly maximised.

The project was conducted between **April 1 and May 14, 2026** at Sabanci University, Tuzla, Istanbul, Turkey. Data was collected from an Apple Watch (physiological sleep metrics), a structured daily self-log (lifestyle variables), and environmental sources (Istanbul 30-year climate normals and astronomically computed sunrise/sunset times for the Tuzla location).

Data Collection & Preprocessing

Data Sources

Table 1: Overview of data sources.

Source	Features	Collection Method
Apple Watch	Sleep stages (Deep / REM / Core / Awake), total duration, bedtime, wake time, Apple sleep score, HRV, SpO ₂ , resting HR, respiratory rate, daily steps, active calories, exercise minutes	Apple Health XML export
Daily Self-Log	Morning feel (0–10), daily productivity (0–10), caffeine intake (mg + last intake time), alcohol (0–5 scale), last meal time	Manual log — mornings & evenings
Weather	Nighttime temperature (°C), humidity (%), barometric pressure (hPa), precipitation (mm)	Istanbul 30-year climate normals (WMO 1991–2021)*
Astronomical	Sunrise time, sunset time, daylight hours, bedtime-after-sunset offset	astral Python library (40.89°N, 29.38°E)

*Direct API access was unavailable in the project environment. Weather features use published WMO 30-year climatological normals — the standard scientific reference for local climate. Sunrise/sunset times are astronomically exact.

Dataset Characteristics

- Total nights observed: **44** (April 1 – May 14, 2026)
- Nights excluded (Apple Watch not worn): **7**
- Complete observations for ML: **37**
- Total features after preprocessing: **24**
- HRV, SpO₂, and resting HR missing for the first 5 nights (Apple Watch calibration period); handled via column-median imputation

Preprocessing

Apple Health XML data was parsed using Python’s `xml.etree.ElementTree` library. Sleep sessions were reconstructed by grouping consecutive stage records, merging gaps of ≤ 45 minutes into the same session. A fragmented April 4 session — where I woke briefly to scroll social media before returning to sleep — was merged into one extended session.

Caffeine was standardised to milligrams using confirmed Starbucks nutritional values: Venti (591 mL) Iced Americano = 300 mg (+75 mg if extra shot); Venti Iced Latte = 225 mg; Grande (483 mL) Iced Latte = 150 mg; Turkish coffee = 60 mg; tea = 40 mg per cup. Alcohol was mapped to a 0–5 scale (0 = none; 2 = 2.5 L beer; 5 = heavy). All remaining NaN values were imputed via column median before training. Features were standardised (zero mean, unit variance) for linear

models; tree-based models used the raw imputed matrix.

Exploratory Data Analysis

Sleep Schedule

My sleep schedule is highly irregular across the 44-night period. Bedtimes ranged from approximately 01:19 to 07:15 with a standard deviation exceeding two hours. Mean sleep duration was approximately 8.5 hours with high variability. Sleep efficiency (time asleep divided by time in bed) averaged approximately 89%.

Apple's proprietary sleep score diverged considerably from my derived efficiency metric on fragmented nights. The starkest example is April 12, where my derived efficiency was 64.5% but Apple's score was 14/100 — a night where 825 mg of caffeine and sustained environmental noise (roommate snoring) severely disrupted sleep architecture.

Target Variables

Both targets used the full 0–10 range, reflecting high sensitivity to nightly conditions. The most prominent anomaly is **April 12**: morning feel = 0 alongside productivity = 10. This is attributable to extreme caffeine intake (825 mg) masking severely disrupted sleep. This single observation creates a structurally contradictory training signal and is documented as a known limitation.

Contrary to the initial hypothesis, a Spearman correlation between the two targets was **not statistically significant** ($r = -0.11$, $p = 0.508$), indicating that morning feel alone does not reliably predict daily productivity within this dataset. This is likely driven by the April 12 anomaly and the broader decoupling of perceived recovery from actual output on high-caffeine days — a key finding that motivates predicting the two targets independently rather than treating one as a proxy for the other.

Caffeine Profile

My daily caffeine intake exceeded the FDA recommended limit of 400 mg on the majority of days, with peaks at 750 mg and 825 mg. The time of last caffeine intake was consistently late (19:00–23:00), well beyond the recommended cutoff of approximately 14:00. This pattern likely contributes to the elevated awake time observed during sleep on many nights.

Hypothesis Tests

Non-parametric tests were used throughout, as the sample size ($n = 37$) precludes normality assumptions. All results are exploratory.

Table 2: Hypothesis test results (Spearman & Mann–Whitney U, $n = 37$).

Hypothesis	Test	Statistic	p -value	Result
Sleep efficiency $\rightarrow$ morning feel	Spearman	$r = +0.709$	<0.001	Significant
Apple sleep score $\rightarrow$ morning feel	Spearman	$r = +0.666$	<0.001	Significant
More REM $\rightarrow$ higher morning feel	Spearman	$r = +0.459$	0.004	Significant
Awake time $\rightarrow$ lower morning feel	Spearman	$r = -0.712$	<0.001	Significant
Higher caffeine $\rightarrow$ lower sleep efficiency	Spearman	$r = -0.377$	0.022	Significant
Later caffeine $\rightarrow$ more awake time	Spearman	$r = +0.328$	0.055	Not significant
Later bedtime (sunset offset) $\rightarrow$ lower feel	Spearman	$r = -0.030$	0.859	Not significant
Morning feel $\rightarrow$ productivity	Spearman	$r = -0.112$	0.508	Not significant
Caffeine ≥ 400 mg vs. < 400 mg	Mann–Whitney	$U=68$	0.002	Significant
Alcohol vs. none: morning feel	Mann–Whitney	$U=56$	0.175	Not significant

Note: With $n = 37$, these results are exploratory. p -values are two-tailed and should not be interpreted as confirmatory evidence.

The strongest signals were **sleep efficiency** ($r = +0.709$), **awake time** ($r = -0.712$), and **Apple sleep score** ($r = +0.666$), all highly significant ($p < 0.001$). High-caffeine nights (≥ 400 mg) produced substantially lower morning feel ratings (mean 4.93 vs. 7.68, $p = 0.002$). Notably, the correlation between morning feel and daily productivity was **not significant** ($r = -0.11$, $p = 0.508$), indicating that within this dataset perceived recovery does not reliably translate into productive output — most likely driven by high-caffeine days where I felt terrible yet remained productive. This motivates predicting the two targets independently rather than treating one as a proxy for the other.

Machine Learning Models

Methodology

Five regression models were trained to predict each target independently, evaluated with **Leave-One-Out Cross-Validation (LOOCV)** — the standard approach for very small datasets. A naive mean baseline (predicting the training-fold mean for every test observation) established

the minimum performance threshold.

Primary models — **Ridge**, **Lasso**, and **ElasticNet** — were chosen for their regularisation properties, appropriate for $n = 37$, following evaluator recommendation. **Random Forest** and **XGBoost** served as secondary comparisons.

Feature Set

Table 3: The 24 features used for ML training.

Category	Features
Sleep stages	duration_h, deep_min, rem_min, core_min, awake_min
Sleep quality	sleep_efficiency, apple_sleep_score, nap_duration_min
Physiological	hrv_ms, spo2_pct, resting_hr, resp_rate
Activity	steps, active_cal, exercise_min
Lifestyle	caffeine_mg, last_caffeine_hour, alcohol_scale, last_meal_hour
Environmental	temp_c, humidity_pct, pressure_hpa, daylight_h, bedtime_after_sunset

LOOCV Results

Naive Baseline

Table 4: Naive mean baseline performance.

Target	Baseline MAE	Baseline R^2
Morning Feel	2.290	−0.056
Productivity	2.793	−0.056

Morning Feel Rating

Table 5: LOOCV results for morning feel rating ($n = 37$). Primary models marked with †; best result highlighted.

Model	MAE	RMSE	R^2	vs. Baseline MAE
Ridge†	1.866	2.214	0.326	−0.424
Lasso† (best)	1.600	1.918	0.494	−0.690
ElasticNet†	1.680	1.969	0.467	−0.610
Random Forest	1.622	2.006	0.447	−0.668
XGBoost	1.654	2.080	0.405	−0.636

Daily Productivity Rating

Table 6: LOOCV results for daily productivity rating ($n = 37$). Primary models marked with †; best result highlighted.

Model	MAE	RMSE	R^2	vs. Baseline MAE
Ridge†	2.889	3.694	−0.317	+0.097
Lasso†	2.565	3.102	0.072	−0.228
ElasticNet†	2.618	3.185	0.021	−0.175
Random Forest	2.548	3.001	0.131	−0.245
XGBoost (best)	2.311	2.724	0.284	−0.481

Key Observations

- **Morning feel is well-predicted.** Lasso achieves $R^2 = 0.494$ under LOOCV, explaining $\approx 49\%$ of variance in a noisy, self-reported target at $n = 37$. All five models substantially beat the naive baseline, with MAE reductions of 0.42–0.69 points. This is a genuinely strong result given the sample size.
- **Productivity is harder to predict.** Ridge actually underperforms the naive baseline (vs. baseline MAE = +0.097), indicating that heavy regularisation overcorrects for this noisier target. XGBoost achieves the best $R^2 = 0.284$, outperforming all primary linear models — a noteworthy exception to the general guidance favouring linear models for small samples, and evidence that daily productivity has non-linear structure.
- **Best primary model:** Lasso for both targets. **Best overall:** Lasso for morning feel ($R^2 = 0.494$); XGBoost for productivity ($R^2 = 0.284$).

SHAP Feature Importance

SHAP values were computed for the best primary model (Lasso) fitted on the full dataset for each target:

- **Morning feel:** Apple sleep score, sleep efficiency, REM duration, and HRV are the strongest positive predictors. High caffeine and late meal timing show consistent negative impacts. Bedtime-after-sunset offset is negatively associated, reflecting circadian misalignment.
- **Productivity:** Despite the non-significant Spearman correlation between the two targets, SHAP analysis on the Lasso productivity model identifies morning feel as its strongest predictor within the trained model — reflecting that the model learned an association even if the raw correlation is weak. Daily steps and exercise minutes show positive associations. Alcohol and high awake time negatively impact productivity.
- **Temperature:** A mild negative association with sleep efficiency appears for both targets, consistent with the known effect of warm nighttime temperatures on sleep architecture.

Personalized Optimal Daily Routine

The following recommendations were derived from the top third of nights ranked by joint composite score (arithmetic mean of normalised LOOCV predictions for morning feel and productivity). These represent observed patterns in my best-performing nights and are not causally validated prescriptions.

Table 7: Personalised optimal routine derived from top-third nights by composite score.

Condition	Optimal (top-third nights)	Overall Median
Sleep Duration	≈6.9 h	≈8.0 h
Bedtime	03:52	03:52
Wake Time	11:03	11:37
Daily Caffeine	≤320 mg	≈340 mg
Last Caffeine Intake	By 16:30	≈18:00
Last Meal	By 19:00	≈20:00
Alcohol	0 (none)	0
Daily Steps	≥21,000	≈15,541
Exercise	≥43 min/day	≈32 min/day
Nap Duration	0 min (none)	0 min
Nighttime Temperature	≈16.5 °C	≈16.3 °C

Summary: Sleep ≈6.9 h, bedtime ≈03:52, last caffeine by 16:30, last meal by 19:00, avoid alcohol, and target ≥21,000 daily steps with ≥43 min exercise.

Note: The `bedtime_after_sunset` feature contains a midnight-crossing artifact: bedtimes after midnight yield negative offset values relative to the previous evening’s sunset because midnight crossing was not accounted for in feature engineering. This limits interpretability of that specific feature; it is excluded from the table above.

Limitations

- **Sample size** ($n = 37$). The dataset is personal and small. All findings are specific to me and cannot be generalised. Statistical power is low; the study is exploratory, not confirmatory.
- **Self-reporting bias.** Morning feel and productivity ratings are subjective and may reflect anchoring effects, mood contamination, or calibration drift across the 44-day collection period.
- **Environmental noise confound.** Elevated `awake_min` on several nights is attributable to roommate snoring rather than my own controllable lifestyle choices, attenuating caffeine- and alcohol-related signals.
- **Weather data.** Direct weather API access was unavailable in the project environment. Weather features are based on published WMO 30-year climate normals (1991–2021) for Istanbul/Tuzla — the standard scientific reference for regional climate — rather than actual observed readings for April–May 2026. Sunrise and sunset times are astronomically exact.
- **Missing Watch nights** ($n = 7$). Nights where I forgot to wear the Apple Watch were excluded from ML training, potentially introducing selection bias.
- **Bedtime-after-sunset artifact.** Bedtimes after midnight produce negative offset values because midnight crossing was not handled in feature engineering.
- **Causality.** All associations are correlational; the model identifies statistical patterns, not causal mechanisms.
- **April 12 anomaly.** The night with morning feel = 0 and productivity = 10 (825 mg caffeine masking severely disrupted sleep) is a structurally contradictory observation that likely biases model coefficients.

Conclusion

I collected 44 nights of data combining Apple Watch sensor output, structured daily self-logging, and environmental enrichment, then trained a dual-target regression pipeline evaluated with LOOCV.

The most significant finding is that morning feel is well-predicted by regularised linear models: Lasso achieves $R^2 = 0.494$ at $n = 37$, explaining approximately half the variance in a noisy self-reported target. Productivity is harder to predict; XGBoost outperforms all models for that target ($R^2 = 0.284$), while Ridge underperforms the naive baseline — highlighting that daily output is driven by non-linear factors that simple regularised models cannot adequately capture.

SHAP analysis identified Apple sleep score, sleep efficiency, REM duration, HRV, and caffeine timing as the features with the greatest personal impact on morning feel. The joint composite optimisation yielded a data-driven routine recommendation: sleep ≈ 6.9 h, last caffeine by 16:30, last meal by 19:00, avoid alcohol, and maintain high physical activity ($\geq 21,000$ steps and ≥ 43 min

exercise daily).

Future work could extend the collection period, incorporate a structured stress metric, integrate Apple Screen Time data, correct the bedtime-after-sunset feature for midnight crossing, and explore multi-output regression to jointly optimise both targets in a single model.